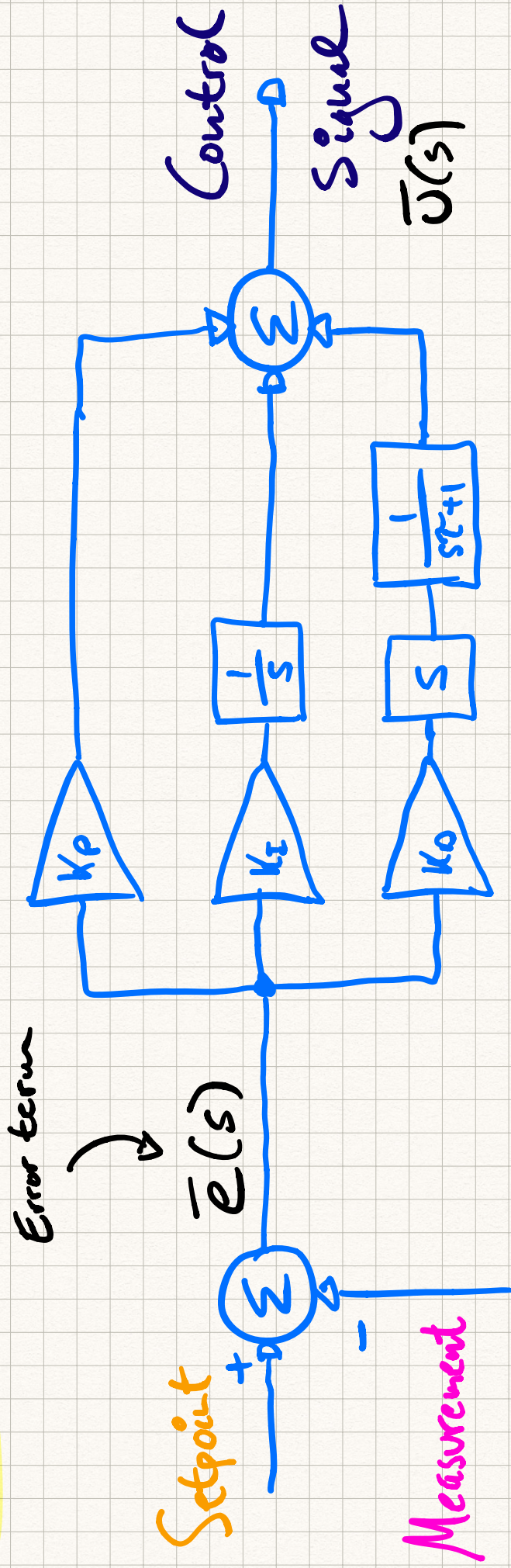


PID in Continuous Domain



$$G(s) = \frac{\bar{u}(s)}{\bar{e}(s)} = \underbrace{K_P}_{\text{Proportional}} + \underbrace{K_I \cdot \frac{1}{s}}_{\text{Integral}} + \underbrace{K_D \cdot \frac{s}{s^2 + 1}}_{\text{(Filtered) Derivative}}$$

After a lot of substituting and rearranging...

$$\begin{bmatrix} p \\ i \\ d \end{bmatrix} = \begin{bmatrix} K_p \cdot e[n] \\ \frac{K_I T}{2} (e[n] + e[n-1]) + i[n-1] \\ \frac{2K_D}{2\tau + T} (e[n] - e[n-1]) + \frac{2\tau - T}{2\tau + T} d[n-1] \end{bmatrix}$$

$$\Rightarrow u[n] = p[n] + i[n] + d[n]$$

↖ Controller output